



Course: BTech

Semester: 7

Prerequisite: Basics of Python & Machine Learning

Course Objective: Natural language processing (NLP) is a crucial part of artificial intelligence (AI). This course will introduce about the core concepts related to Natural Language Processing and its applications. A broad introduction about incorporation of Deep Learning techniques in Natural Language Processing is considered.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	0	0	0	3	20	20	-	60	-	100

SEE - Semester End Examination, T - Theory, P - Practical

Course Content

W - Weightage (%) , T - Teaching hours

Sr.	Topics	W	T
1	Introduction to Natural Language Processing: The First Wave: Rationalism, The Second Wave: Empiricism, The Third Wave: Deep Learning, what makes Natural Language Processing Difficult? What do we want to achieve through Natural Language Processing? Common Terms Associated with Language Processing, Introduction to basics of NLP operations: Word Level Analysis, Syntactic Analysis, Semantic Analysis, Word Sense Disambiguation, Discourse Processing, Part of Speech (PoS) Tagging, Natural Language Inception, Information Retrieval, Applications of NLP.	15	8
2	Introduction to Deep Learning: Introduction to the Deep Learning, Basic Structure of Neural Network, Types of Activation Function, Types of Neural Network: Feedforward Neural Network, Convolutional Neural Network, Recurrent Neural Network, Encoder Decoder Networks, Recursive Neural Network, Multilayer Neural Network, Stochastic Gradient Descent, Backpropagation Algorithm.	15	8
3	Word Vector Representations: Introduction to Word Embedding: Neural Language Model, Word2Vec: Skip-Gram Model, Model Components: Architecture, Hidden Layer, Output Layer, CBOW Model, Subsampling Frequent Words: Negative Sampling. Efficient Softmax approximations, Sampling-based approximations for Softmax, Glove, Cross-lingual word embedding models	20	10
4	Recurrent Neural Network for NLP: What is RNN? Recurrence, Difference between Feedforward and Recurrent Neural Networks, Basics of Recurrent Neural Networks, Natural Language Processing & Recurrent Neural Networks, RNN's Mechanism, Training RNN, Meta meaning of Hidden State of RNN. Tuning RNN, Long Short-Term Memory Networks, Sequence to Sequence Models, Advance Sequence to Sequence Models.	20	8
5	Chatbot Development: Introduction to the Chatbot, Origin of Chatbot, How does a chatbot work? Why are chatbots such a big opportunity, Conversational bot, Chatbot: Automatic Text Generation procedure, Introduction to ChatGPT AI Chatbot	10	4
6	Deep Learning in Sentiment Analysis: Introduction, Sentiment- Specific Word Embedding, Sentence Level Sentiment Classification: Convolution Neural Network, Recurrent Neural Network, Recursive Neural Network, Integration of External Resources, Document Level Sentiment Classification, Fine-Grained Sentiment Analysis, Opinion Mining, Targeted Sentiment Analysis, Aspect Level Sentiment Analysis, Stance Detection, Sarcasm Recognition	20	7



Reference Books

1.	Deep Learning for Natural Language Processing: Creating Neural Networks with Python: P. Goyal, S. Pandey and K. Jain by Apress. (TextBook)
2.	Deep Learning in Natural Language Processing: Li Deng, Yang Liu by Springer.
3.	Natural Language Processing: Develop Deep Learning Models for Natural Language in Python by Jason Brownlee
4.	Handbook of Natural Language Processing: Nitin Indurkha Fred J. Damerau, CRC Press.

Course Outcome

After Learning the Course the students shall be able to:

1. Understand the concepts of Natural Language Processing.
2. Understand the concepts of Deep Learning
3. Apply several models of Deep Learning in Natural Language Processing
4. Analyze the importance of Recurrent Neural Network & LSTM in NLP.
5. Understand about recent developments in the field of Chatbots.
6. Classify and Implement Sentimental Analysis techniques.